

Nishant Luhera

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Experience

MershLab(Stealth Startup)

May 2026 – Present

Founding Engineer

Gurgaon & Remote

- Architected, end-to-end, an AI-driven WhatsApp lead-acquisition platform for Indian SMBs (founding engineer, 3-person team): a **4-package TypeScript monorepo** (Next.js 14 / React, Node.js / Express) with a custom **DAG workflow engine of 66 reusable nodes** on a multi-tenant SQLite (better-sqlite3, WAL) data layer.
- Orchestrated an event-driven backend with an in-process job queue providing **per-tenant round-robin fairness**, automatic retries, and idempotent webhook processing, ingesting leads from IndiaMART webhooks, Click-to-WhatsApp, and CSV import.
- Integrated a **provider-agnostic AI gateway** abstracting Anthropic Claude and DeepSeek behind a config-driven two-tier (quality/fast) interface, with per-tenant token metering, prompt caching, and bring-your-own-key (BYOK) support.
- Secured the platform with AES-256-GCM credential encryption, JWT auth, signature-verified webhooks, SSRF protection, rate limiting, and a DPDP-aware tenant-isolation model; wired in the Meta WhatsApp Cloud API via a pluggable transport-seam abstraction.

Livepeer

Oct 2025 – Nov 2025

Open-Source Contributor (Paid Bounty)

Remote

- Implemented an image **inpainting pipeline** for Livepeer's AI Worker (`ai-worker`), adding mask-guided diffusion image generation to the network's decentralized GPU inference jobs.
- Wrapped a Stable Diffusion inpainting model (HuggingFace `diffusers`) behind the runner's standard pipeline interface — FastAPI route, typed request/response schemas, and a containerized GPU runtime matching the repo's existing pipeline contract.
- Completed and paid as an \$850 bounty; Livepeer's core team carried the contribution forward for upstream integration.

Indian Institute of Technology, Bombay

Mar 2024 – July 2024

Research Intern – NLP

Remote

- Researched **LLM-based approaches for script detection** — automatically identifying the writing script of multilingual text — with a focus on **Indic scripts** (e.g. Devanagari, Bengali, Tamil).
- Designed an evaluation pipeline to **benchmark language models for classification accuracy** across Indic scripts, comparing models and surfacing systematic failure cases on low-resource scripts.

Projects

AtlasInfer | PyTorch, Triton, CUDA, Quantization

🐙 [GitHub](#) 2025 – 2026

- Engineered a from-scratch mixed-precision LLM quantization engine (per-layer FP16/INT8/4-bit NF4) on PyTorch with no `bitsandbytes/auto-gptq` dependency; matches `bitsandbytes`' INT8 and **beats its NF4 perplexity** on Qwen3-0.6B and Qwen2.5.
- Cast per-layer bit-width assignment as a **multiple-choice knapsack** solved via dynamic programming over a calibration-based sensitivity profiler, cutting 4-bit perplexity loss ~35% at equal memory vs. uniform quantization.
- Wrote fused dequant-GEMM **Triton kernels** (W8A16/W4A16) reaching **1.4–1.95×** faster batch-1 decode than FP16 on an RTX 3060, with an eager fallback for CPU/Windows.

CodeQuorum | Node.js, TypeScript, Yjs (CRDT), Socket.IO, SQLite

🐙 [GitHub](#) 2025 – 2026

- Built a real-time collaborative multi-file IDE where concurrent edits converge **conflict-free** via Yjs CRDTs over Socket.IO, with live cursors, presence, and follow/spotlight modes.
- Enforced authorization **server-side** (private/read-only rooms, dropped viewer edits) with `scrypt`+JWT auth and GitHub OAuth; persisted an authoritative per-room CRDT document to SQLite.
- Shipped collaborative Jupyter notebooks with in-browser Python via a self-hosted Pyodide kernel, plus sandboxed JS/HTML execution (timed Web Worker + locked-down `iframe/CSP`).

ECG Analysis Suite | PyTorch, OpenCV, scikit-learn, Streamlit

🐙 [GitHub](#) 2024 – 2026

- Trained a 1D-CNN beat classifier to **98.5% test accuracy** (MIT-BIH) and a 1D-ResNet ensemble to **0.935 macro-AUROC** on PTB-XL official folds, exceeding the best published single-CNN benchmark.

- Developed an ECG-image-to-diagnosis pipeline (OpenCV signal digitization → calibrated 1D-ResNet → Grad-CAM explanation), deployed as an interactive Streamlit demo.
- Added a clinical operating-point analysis (sensitivity ≥ 0.90 , PPV at realistic prevalence) demonstrating why a high AUROC is not a deployable medical device.

Education

BML Munjal University

2022-2026

B.Tech in Computer Science & Engineering

India

- **Relevant Coursework:** Data Structures and Algorithms (C++), Prob & Stat in CS (Python), Operating Systems, Database, Linear Algebra

Skills

Languages: Python, TypeScript, Go, Rust, C/C++

AI / Machine Learning: PyTorch, TensorFlow, CUDA, Triton, HuggingFace Diffusers, LLM APIs (Anthropic, DeepSeek), Quantization (INT8/NF4), scikit-learn, OpenCV

Infrastructure: Linux/Unix, Docker, Git, CI/CD, SQLite (WAL), Cloud Computing, Multi-tenant Architecture

Backend: Node.js, Express.js, FastAPI, Django, Flask, REST APIs, WebSockets / Socket.IO

Frontend: React.js, Next.js, Tailwind CSS, HTML5/CSS3